

Chapter **A2**

Agate Ltd Case Study— Requirements Model

Agate Ltd

A2.1 | Introduction

In this chapter we bring together the models (diagrams and supporting textual information) that constitute the requirements model. In Chapters 5 and 6 we have introduced the following UML diagrams:

- use case diagram
- activity diagram
- package diagram.

There is not the space in this book to produce a complete requirements model. However, in this chapter we have included a sample of the diagrams and other information. This is done to illustrate the kind of material that should be brought together in a requirements model. We have also tried to illustrate how iteration of the model will produce versions of the model that are elaborated with more detail.

A2.2 | Requirements List

The requirements list on the next page includes a column to show which use cases provide the functionality of each requirement. This requirements list includes some use cases not in the first iteration of the use case model.

No.	Requirement	Use Case(s)
1	To record names, address and contact details for each client	Add a new client
2	To record the details of each campaign for each client. This will include the title of the campaign, planned start and finish dates, estimated costs, budgets, actual costs and dates, and the current state of completion	Add a new campaign
3	To provide information that can be used in the separate accounts system for invoicing clients for campaigns	Record completion of a campaign
4	To record payments for campaigns that are also recorded in the separate accounts system	Record client payment
5	To record which staff are working on which campaigns, including the campaign manager for each campaign	Assign staff to work on a campaign
6	To record which staff are assigned as staff contacts to clients	Assign a staff contact
7	To check on the status of campaigns and whether they are within budget	Check campaign budget
8	To allow creative staff to record notes of ideas for campaigns and adverts (concept notes)	Create concept note
9	To provide other staff with access to these concept notes	Browse concept notes
10	To record details of adverts, including the progress on their production	Add a new advert to a campaign. Record completion of an advert
11	To schedule the dates when adverts will be run	Add a new advert to a campaign
12	To maintain staff records for creative and administrative staff	Add a new member of staff
13	To maintain details of staff grades and the pay for those grades	Add a new staff grade. Change the rate for a staff grade
14	To record which staff are on which grade	Change the grade for a member of staff
15	To calculate the annual bonus for all staff	Calculate staff bonuses
16	To enable data about clients, campaigns, adverts and staff to be shared between offices	Not applicable
17	To allow the system to be modified to work in different languages	Not applicable
18	To restrict the ability to create or update data to authorized users in the company.	All use cases that create or update data
19	To limit planned downtime to one hour a week during the night UK time.	Not applicable

A2.3 Actors and Use Cases

Actor	Description
Accountant	The accountant works in the Accounts department and is responsible for the major resourcing issues for campaigns including staffing and related financial matters.
Campaign Manager	Either a Director or an Account Manager (job titles), who is responsible for estimating the campaign cost and agreeing it with the client. They are responsible for assigning staff to the team and supervising their work, managing the progress of the campaign, conducting any further budget negotiations and authorizing the final invoices.
Staff Contact	Member of staff who is the contact for a particular client. They provide a first point of contact for the client when the client wants to contact Agate.
Staff	Any member of staff in Agate.
Campaign Staff	Member of staff working on a particular campaign.

Figures A2.1 to A2.3 show the use cases from the first iteration, with use case descriptions in the tables. The use case diagram in Figure A2.1 has been drawn in a modelling tool, Enterprise Architect from SparxSystems, and is shown as a screenshot.

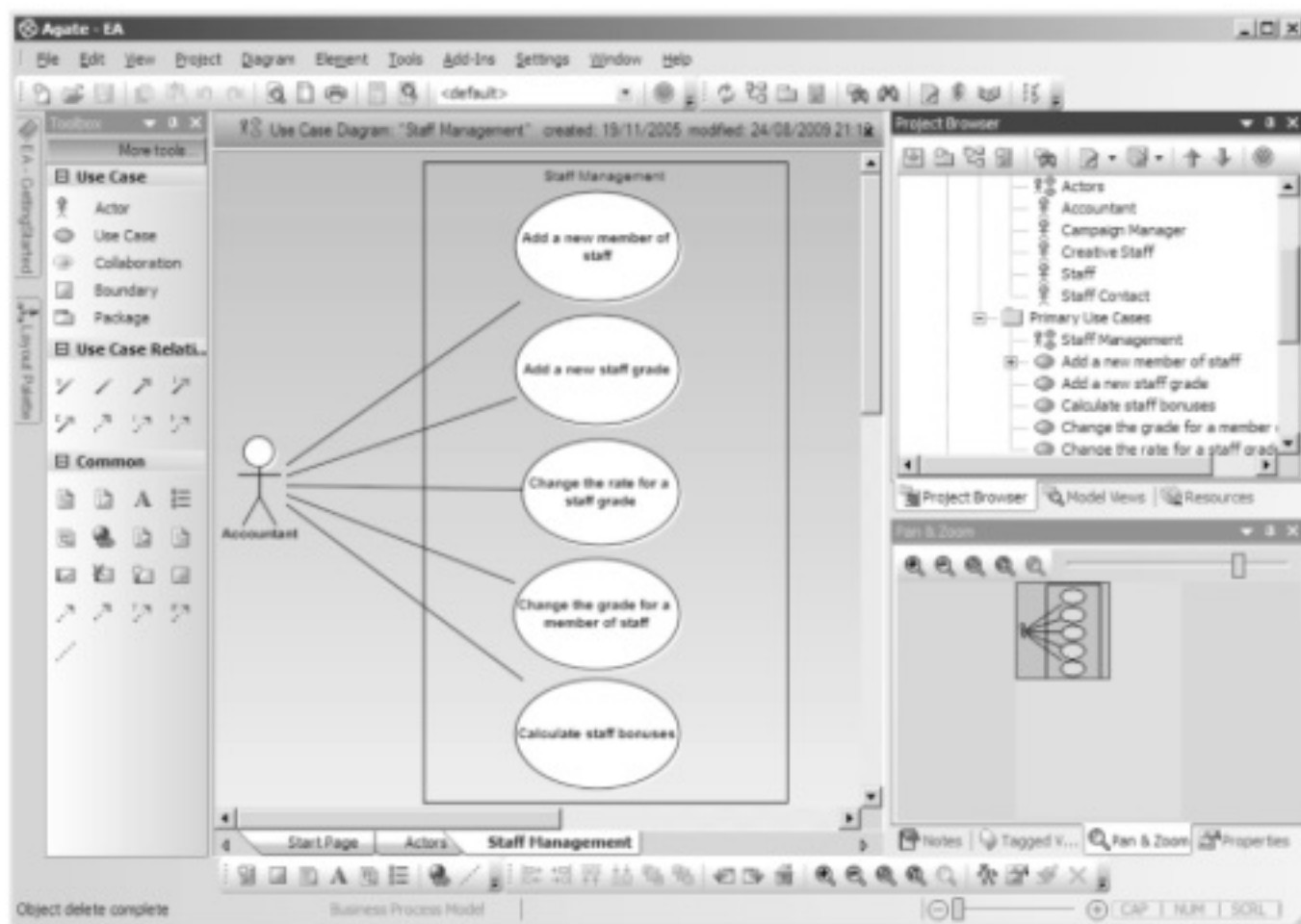


Figure A2.1 Staff Management use cases.

Use Case	Description
Add a new staff member	When a new member of staff joins Agate, his or her details are recorded. He or she is assigned a staff number, and the start date is entered. Start date defaults to today's date. The starting grade is entered.
Add a new staff grade	Occasionally a new grade for a member of staff must be added. The name of the grade is entered. At the same time, the rate for that grade and the rate start date are entered; the date defaults to today's date.
Change the rate for a staff grade	Annually the rates for grades are changed. The new rate for each grade is entered, and the rate start date set (no default). The old grade rate is retrieved and the rate finish date for that grade rate set to the day before the start of the new rate.
Change the grade for a staff member	When a member of staff is promoted, the new grade and the date on which they start on that grade are entered. The old staff grade is retrieved and the finish date set to the day before the start of the new grade.
Calculate staff bonuses	At the end of each month staff bonuses are calculated. This involves calculating the bonus due on each campaign a member of staff is working on. These are summed to give the total staff bonus.

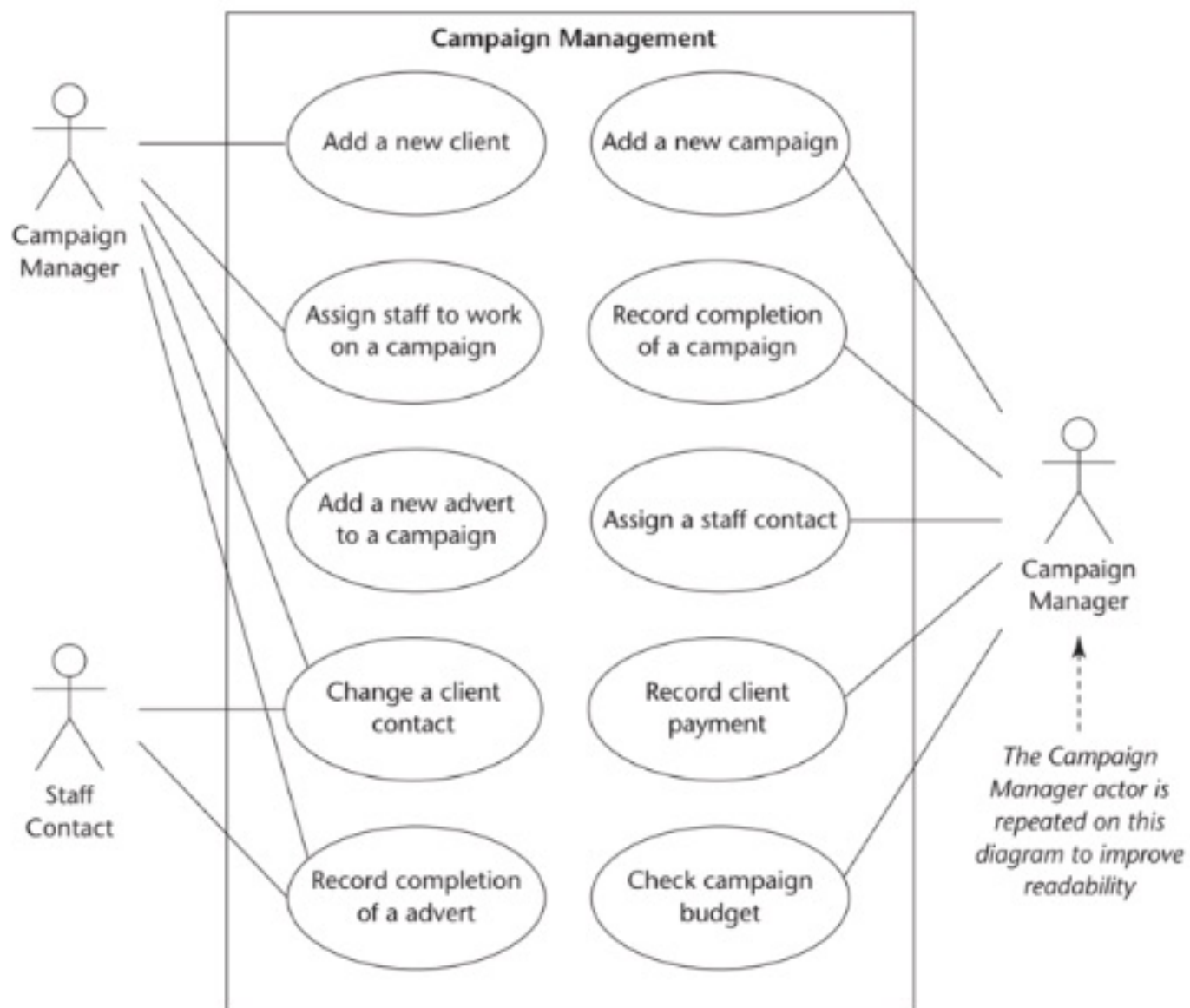


Figure A2.2 Campaign Management use cases.

Use Case	Description
Add a new client	When Agate obtains a new client, the full details of the client are entered. Typically this will be because of a new campaign, and therefore the new campaign will be added straight away.
Assign staff to work on a campaign	The campaign manager selects a particular campaign. A list of staff not already working on that campaign is displayed, and he or she selects those to be assigned to this campaign.
Add a new advert to a campaign	A campaign can consist of many adverts. Details of each advert are entered into the system with a target completion date and estimated cost.
Change a client contact	Records when the client's contact person with Agate is changed.
Record completion of an advert	The actor selects the relevant client, campaign and advert. The selected advert is then completed by setting its completion date.
Add a new campaign	When Agate gets the business for a new campaign, details of the campaign are entered, including the intended finish date and the estimated cost. The manager for that campaign is the person who enters it.
Record completion of a campaign	When a campaign is completed, the actual completion date and cost are entered. A record of completion form is printed out for the Accountant as the basis for invoicing the client.
Assign a staff contact	Clients have a member of staff assigned to them as their particular contact person.
Record client payment	When a client pays for a campaign, the payment amount is checked against the actual cost and the date paid is entered.
Check campaign budget	The campaign budget may be checked to ensure that it has not been exceeded. The current campaign cost is determined by the total cost of all the adverts and the campaign overheads.

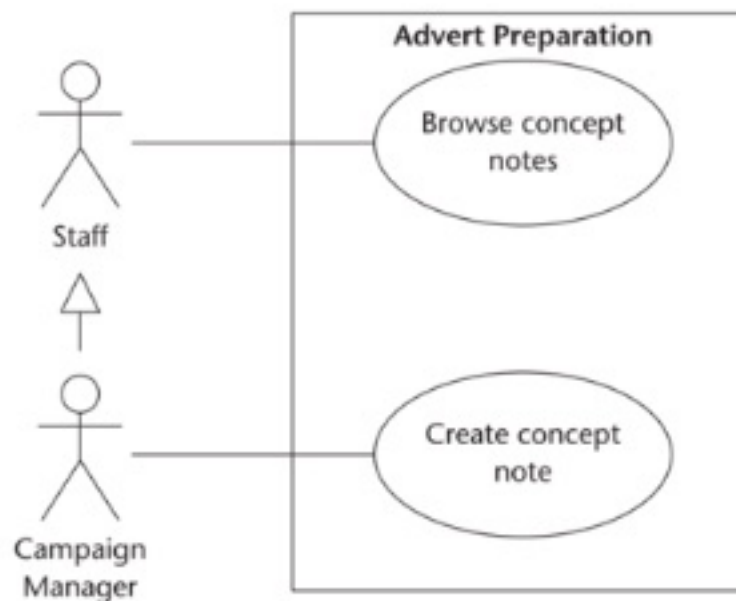


Figure A2.3 Advert Preparation use cases.

Use Case	Description
Browse concept notes	Any member of staff may view concept notes for a campaign. The campaign must be selected first. The titles of all notes associated with that campaign will be displayed. The user will be able to select a note and view the text on screen. Having viewed one note, others can be selected and viewed.
Create concept note	A member of staff working on a campaign can create a concept note, which records ideas, concepts and themes that will be used in an advertising campaign. The note is in text form. Each note has a title. The person who created the note, the date and time are also recorded.

As part of the second iteration of use case modelling, it is suggested that all the use cases that require the user to select a client, a campaign or an advert should have include relationships with use cases called *Find client*, *Find campaign* and *Find advert*. An example of this is shown in Fig. A2.4.

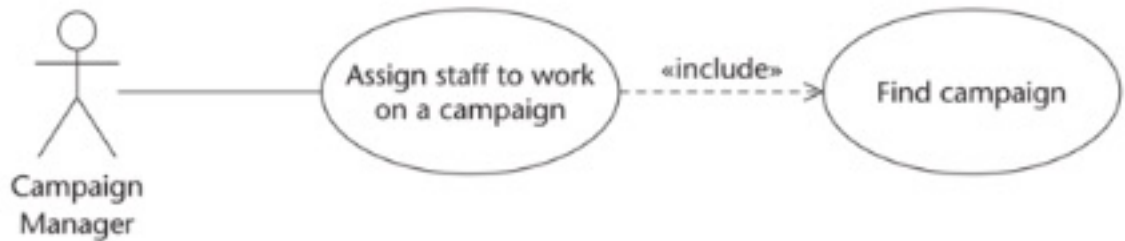


Figure A2.4 Inclusion of *Find campaign* use case.

In order to test out this idea, prototypes of the user interface were produced in the second iteration. The first prototypes used a separate user interface for these included use cases, as shown in Fig. A2.5.

However, feedback from the users indicated that this approach was not acceptable. They did not want to have to keep opening extra windows to find clients, campaigns and adverts. The users expressed the view that they should be able to select these from



Figure A2.5 Prototype interface for the *Find campaign* use case.

listboxes or dropdown lists that were part of the interface for whatever use case they were in at the time.

In the third iteration of use case modelling, a set of prototypes was produced that uses listboxes. Figure A2.6 shows an example.

In the third iteration, some additional functionality was identified and added to the use case diagrams. As an example of this, Fig. A2.7 shows the use case *Check campaign budget* extended by the use case *Print campaign summary* and *Print campaign invoice*. This additional functionality will also require a change to the prototype interface in Fig. A2.6. Two additional buttons, *Print Summary* and *Print Invoice*, need to be added to the row of buttons at the bottom of the window.

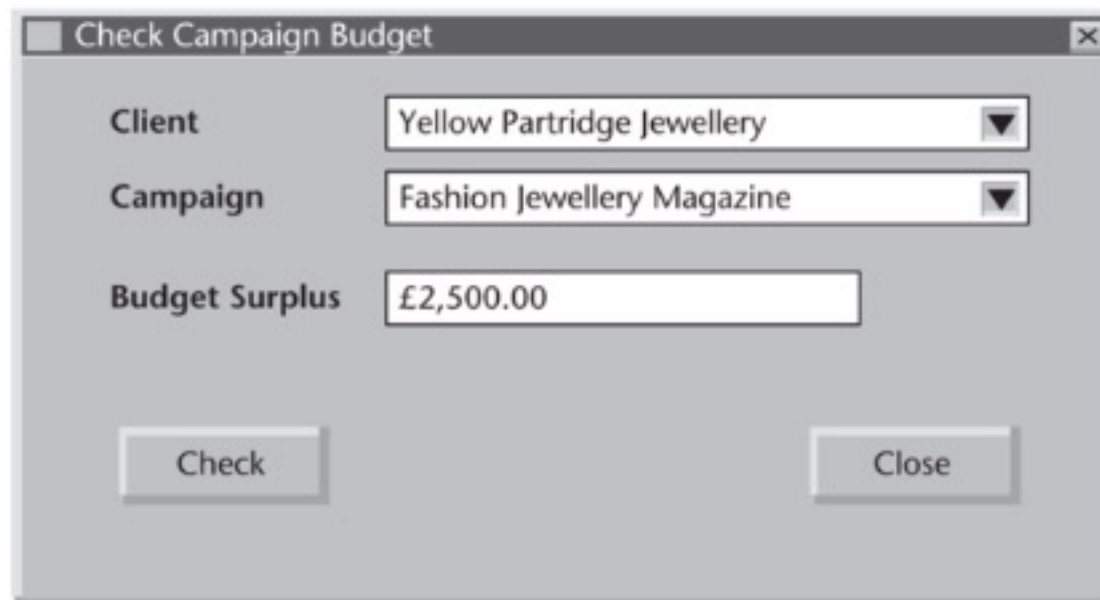


Figure A2.6 Prototype interface for the use case *Check campaign budget*.

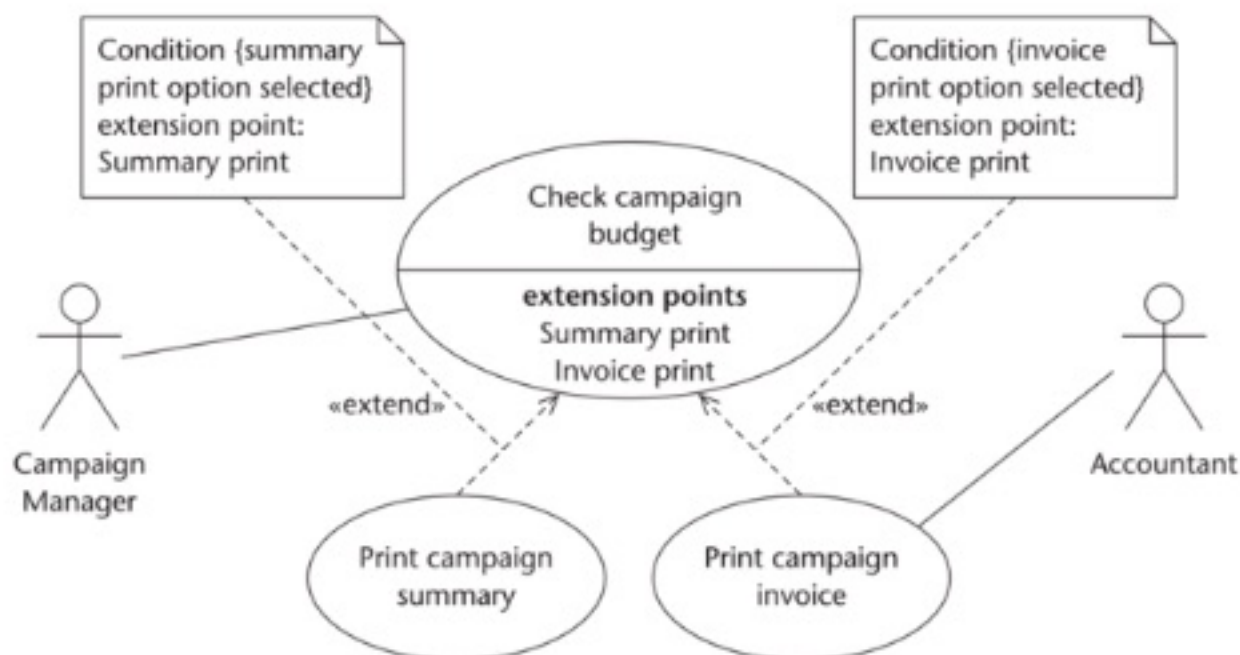


Figure A2.7 Modified use case *Check campaign budget* with extensions.

Also in the third iteration, the use case descriptions are elaborated to provide more detail about the interaction between the actors and the system. Two examples of these use case descriptions are provided below.

Use case description: Check campaign budget

Actor Action	System Response
1. None	2. Lists the names of all clients
3. The actor selects the client name	4. Lists the titles of all campaigns for that client
5. Selects the relevant campaign. Requests budget check	6. Displays the budget surplus for that campaign
Extensions	
After step 6, the campaign manager prints a campaign summary.	
After step 6, the campaign manager prints a campaign invoice.	

Use case description: Assign staff to work on a campaign

Actor Action	System Response
1. None	2. Displays list of client names
3. The actor selects the client name	4. Lists the titles of all campaigns for that client
5. Selects the relevant campaign	6. Displays a list of all staff members not already allocated to this campaign
7. Highlights the staff members to be assigned to this campaign. Clicks Allocate button.	8. Presents a message confirming that staff have been allocated
Alternative Courses	
None.	

A2.4 Glossary

A glossary of terms has been drawn up, which lists the specialist terms that apply to the domain of this project—advertising campaigns.

Term	Description
Admin Staff	Staff within Agate whose role is to provide administrative support that enables the work of the creative staff to take place, for example secretaries, accounts clerks and the office manager
Advert	An advertisement designed by Agate as part of a campaign. Adverts can be for TV, cinema, websites, newspapers, magazines, advertising hoardings, brochures or leaflets. Synonym: Advertisement
Agate	An advertising agency based in Birmingham, UK, but with offices around the world. The customer for this project
Campaign	An advertising campaign. Adverts are organized into campaigns in order to achieve a particular objective, for example a campaign to launch a new product or service, a campaign to rebrand a company or product, or a campaign to promote an existing product in order to take market share from competitors
Campaign Staff	Member of staff working on a particular campaign

Term	Description
Client	A customer of Agate. A company or organization that wishes to obtain the services of Agate to develop and manage an advertising campaign, and design and produce adverts for the campaign
Concept Note	A textual note about an idea for a campaign or advert. This is where creative staff record their ideas during the process of deciding the themes of campaigns and adverts. Synonym: Note
Creative Staff	Staff with a creative role in the company, such as designers, editors and copywriters; those who are engaged in the work of the company to develop and manage campaigns and design and produce adverts
Grade	A job grade. Each member of staff is on a particular grade, for example 'Graphic Artist 2' or 'Copywriter 1'
Grade Rate	The rate of pay for a particular grade, for example the Grade 'Graphic Artist 2' is paid £26 170 per year in the UK from 1/1/2010 to 31/12/2010
Staff	Any member of staff in Agate. Synonyms: Staff member, member of staff

A2.5 Initial Architecture

The initial architecture of the system is based on the packages into which the use cases are grouped. These use cases have been grouped into three subsystem packages: Campaign Management, Staff Management and Advert Preparation.

Figure A2.8 shows the initial architecture of these three packages, and a package that will provide the mechanisms for the distribution of the application. At this early stage in the project, it is not clear what this will be, but something will be necessary to meet Requirement 16. At this stage the packages have names that reflect the business context rather than how they might be implemented in Java packages or C# namespaces or an equivalent structure. This will change later.

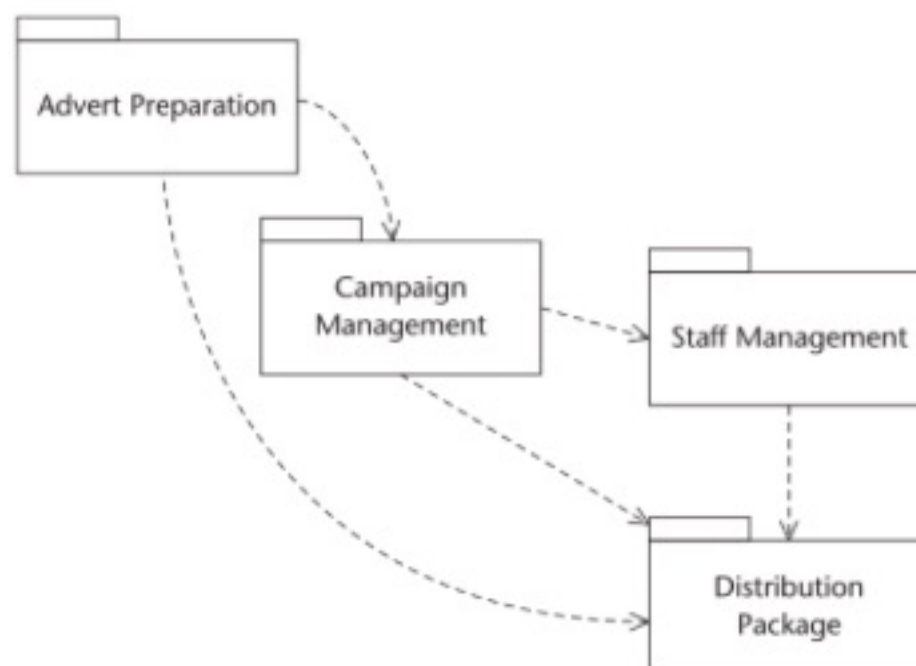


Figure A2.8 Initial package architecture.

A2.6 Activities of Requirements Modelling

In Chapter 5, we outlined the phases and activities of the iterative lifecycle, and in Chapter 6 we included an activity diagram to show the activity `Requirements capture and modelling`. Figure A2.9 shows the same diagram. This activity can be broken down into other activities, and these are shown in Figs A2.10, A2.11 and A2.12.

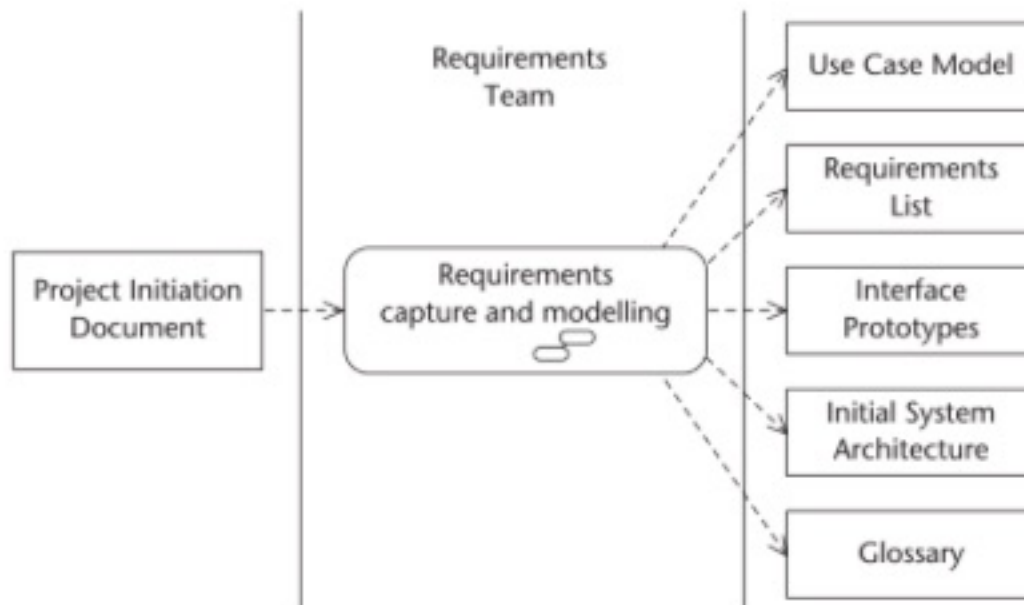


Figure A2.9 Activity diagram for `Requirements capture and modelling`.

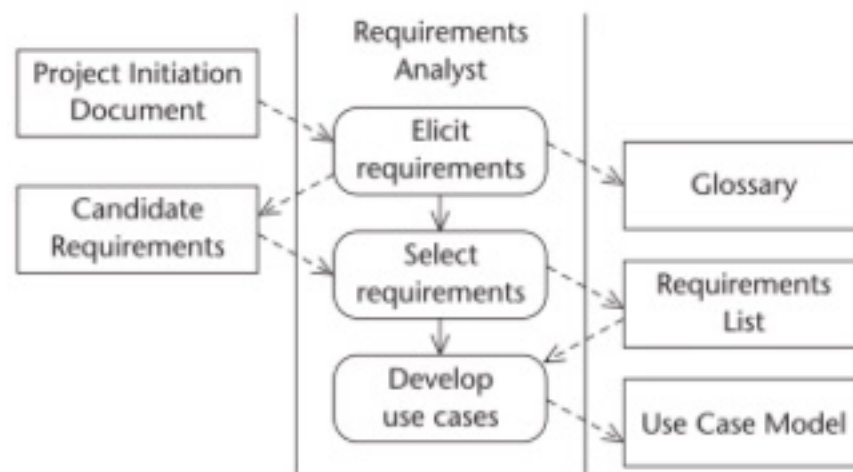


Figure A2.10 Activity diagram to show the activities involved in capturing requirements.

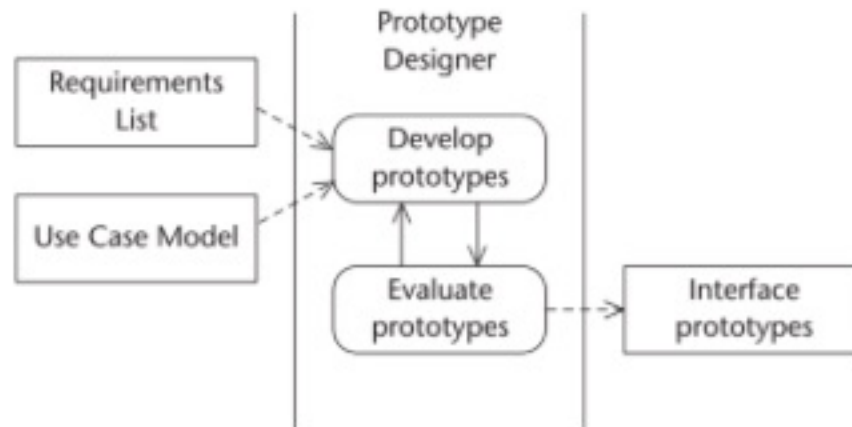


Figure A2.11 Activity diagram to show the activities involved in developing prototypes.

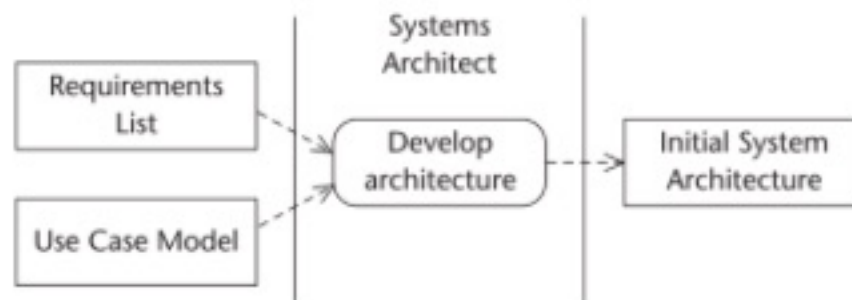


Figure A2.12 Activity diagram to show the activities in developing an initial architecture.

It is important to remember that in a project that adopts an iterative lifecycle, these activities may take place over a series of iterations. In the first iteration, the emphasis will be on requirements capture and modelling; in the second, it will shift to analysis, but some requirements capture and modelling activities may still take place. Refer back to Fig. 5.15. You may also want to look at Fig. 5.7, which illustrates the development of the use case model through successive iterations.